Damla Görgülü

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Education

University of Cambridge, Cambridge, United Kingdom 10/2026 – Present
MPhil in Electrical and Electronic Engineering

- **Michaelmas modules:** Photonic Devices; Very Large Scale Integration (VLSI); Electric Vehicles and Transport Electrification; Integrated Electronic Devices.
- **Lent Term Modules:** Quantum Computing, Communications and Emerging Technologies; Entrepreneurship, University Spin-offs and IP Management; Sensor, Embedded and Control Systems; Bioelectronics.
- **Scholarship:** Turkish Education Foundation (UK) Cambridge Masters Scholarship (Cambridge Trust).

Koç University, İstanbul, Turkey
Bachelor of Computer Engineering (Double Major), GPA 3.87/4.00 09/2022 – 02/2026
Bachelor of Electrical and Electronics Engineering, GPA 3.87/4.00 09/2021 – 07/2025

- Completed a specialisation in Artificial Intelligence and graduated summa cum laude in both Electrical and Electronics Engineering and Computer Engineering.
- **Awards:** Dean's Honour Roll; Vehbi Koç Honour Award (awarded for semester GPAs above 3.75; received for eight consecutive semesters).

Publications

- **D. Görgülü**, F. A. Güler, A. B. Trabzon and M. C. Onbaşlı, "Benchmarking Skyrmion Logic Against Foundry CMOS Circuits" (submitted to Nanoscale, 2026). The first three authors contributed equally.

Research Experience

Researcher, Koç University — Hypernetwork-Based LoRA Generation 06/2025 – Present

- Investigating **hypernetwork-based LoRA generation** for content–style disentanglement in diffusion models, as part of a four-person team led by Assoc. Prof. Aykut Erdem.
- Evaluating **LoRA fusion strategies, content-preserving datasets and structural loss functions** for composable content and visual style in FLUX-based image synthesis.

Researcher, Koç University — Spintronics and CMOS 02/2025 – 09/2026

- Co-developed an **RTL-to-skyrmion benchmarking workflow** for a 4-bit ALU, 8-point DFT engine and multilayer perceptron, comparing SKY130 CMOS implementations with synthetic-antiferromagnetic skyrmion logic.
- Quantified area, latency and energy trade-offs; mapped skyrmion networks showed **3.44–4.06× smaller core-logic footprints** than CMOS, excluding peripheral circuitry.
- Co-authored a research paper reporting the benchmarking methodology and findings, submitted to Nanoscale (2026).

Researcher, Technical University of Munich, Chair of AI Processor Design 03/2026 – 06/2026

- Designed **weight and bias partitioning** and transfer schedules to improve memory-bandwidth utilisation in neural-network accelerators.
- Implemented **S-Link streaming transfers** between a Cheshire RISC-V SoC and a ButterStick FPGA to support neural-network inference using external DRAM.
- Designed and executed test cases for a **dual-FPGA S-Link setup**, measured data-transfer performance and presented findings to the research supervisor.

Researcher, Technical University of Munich, Chair of AI Processor Design 07/2025 – 12/2025

- Brought up **LiteX-based RISC-V SoCs** on OrangeCrab and ButterStick FPGAs and established a Linux-booted SD-card-to-DRAM path for hardware experiments.

- Integrated custom RTL counters and memory-processing modules using control and status registers (CSRs) to verify SoC memory and I/O interfaces.
- Investigated DRAM retention and refresh behaviour and presented experimental findings in technical reviews.

Researcher, Technical University of Munich, Chair of AI Processor Design 06/2024 – 09/2024

- Investigated **BTI/HCI transistor ageing** under Prof. Hussam Amrouch using the CARAT framework, examining reliability effects on AI processor circuits.
- Adapted the framework to **3 nm and 14 nm** technology nodes, simulated Verilog adders, multipliers and multiply–accumulate units, and analysed ageing-induced delay variation using MATLAB and Synopsys WaveView.

Undergraduate Researcher, Koç University — RISC-V Microprocessor Design 06/2023 – 09/2024

- Co-developed a **RISC-V-based CNN inference processor** for MNIST classification with two peers, implementing digital microprocessor and accelerator modules in **VHDL and Verilog**.
- Quantised CNN weights to Q0.15, integrated UART communications and developed a Python interface for input testing and verification.
- Selected as a finalist in the **TEKNOFEST 2024 Microprocessor Design Competition**.

Computer Vision Intern, Savunma Teknolojileri ve Mühendislik A.Ş. (STM) 07/2023 – 08/2023

- Researched **GAN-based licence-plate super-resolution** and trained Real-ESRGAN on **3,500+ images**, comparing paired and dynamically generated datasets.
- Evaluated **4× and 16×** super-resolution outputs on drone imagery from field tests.

Selected Teaching Experience

Tutor, Koç University KOLT Tutoring Center 02/2024 – 06/2024

- **Digital System Design:** Held weekly office hours for over 120 students, providing guidance on digital logic and circuit design problems.
- **Electromagnetism:** Conducted three weekly office hours and delivered pre-examination review lectures with worked examples.

Teaching Assistant, Koç University — Signals and Systems 02/2023 – 06/2023

- Co-led weekly problem-solving sessions and held office hours for over 50 students as part of a four-person teaching team.
- Prepared course quizzes to assess students' understanding of signals and systems concepts.

Leadership and Activities

Co-Founder and Editor, Koç University Undergraduate Journal of Engineering 09/2023 – 09/2024

- Co-founded an undergraduate engineering journal and contacted students completing research and thesis projects to encourage submissions.
- Edited student research articles and coordinated with contributors to communicate technical work to a wider engineering audience.

Board Member, Koç University IEEE Student Branch 09/2022 – 06/2023

Percussionist, Koç University Orchestra 09/2022 – Present

Skills and Interests

- **Hardware and EDA:** VHDL, Verilog, FPGA/SoC design, RISC-V, LiteX, OpenLane/SKY130, Synopsys WaveView.
- **Programming and Tools:** Python, C/C++, Java, MATLAB, Linux/Unix, Git.
- **Languages:** Turkish (Native), English (C2), Spanish (A2–B1), German (A1).
- **Interests:** Drums (London College of Music Grade 8); sailing (four summers as an instructor); illustration and handcrafts; skiing.